

Analysis of Transistor-level CMOS Inverter in LTspice

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1 Introduction

2 Circuit Setup

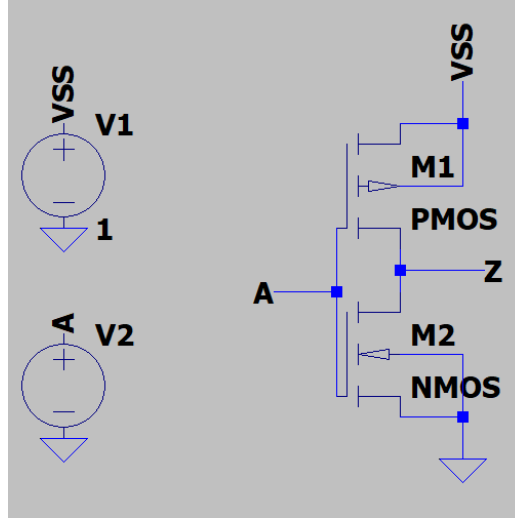


Figure 1: Circuit Setup

From Fig. 1, the setup of the CMOS circuit in LTspice is shown. First, on the left side, there are two voltage sources, V1 represents the power supply voltage (Vdd) and is set to 5V, while V2 represents the input voltage (A). Then, on the right side, there are two MOSFETs used to represent the PMOS and NMOS transistors. The upper side is the PMOS transistor, which is connected to Vdd, while the lower side is the NMOS transistor, which is connected to the ground. Finally, the gate of both transistors is connected to A and the output voltage (Z) is connected to the junction between the two transistors.

[?]

3 Simulation Results

References